Data Mining and Machine Learning (Group Coursework)

F21DL (2026/2027)

In this group coursework, you and your team will work on a data mining and machine learning project.

The project will cover various aspects of data mining and machine learning, including basic concepts, exploration and data analysis, generative models, discriminative learning, and practical application of these techniques. Sometimes we will refer to this project as your "Machine learning portfolio".

The project is designed to give you hands-on experience with real-world data and machine learning algorithms while also teaching you how to effectively collaborate.

Note: Check with your course coordinator on the exact tools you will be using, such as GitHub (or Gitlab in Edinburgh) for version control, documentation, code reviews, regular commits, testing, and data cleaning.

Note: For your convenience, we've underlined the parts of this coursework descriptor that are different from the F20DL version ☺

1. Coursework Objectives

1. Gain a solid grasp of the fundamental concepts in data mining (datasets, pre-processing) and machine learning (unsupervised learning, supervised learning, neural networks, etc.).
2. Explore machine learning methodologies such as naïve Bayes, probabilistic graphical models, and cluster analysis (including k-means clustering and the EM algorithm).
3. Implement discriminative learning methods such as linear models, linear regression, decision tree learning, perceptron, ensemble methods and advanced models like multi-layer perceptron and deep learning architectures.
4. Develop practical skills by working on real-world datasets, conducting data preprocessing, implementing machine learning models, and evaluating their performance.
5. (**Campus-dependent**) Learn how to effectively collaborate with your team members using GitHub (or Gitlab in Edinburgh) for version control, code sharing, and project management. Your coursework documentation and evidence should be hosted on GitHub/Gitlab.

2. Summary of the Group Project

- Each group will select a broad topic of application, some examples are provided in Section 3.1 – Topics, below.
- Then they should look for publicly available real-world datasets (at least one and at most three), suitable for the topic of choice and applicable model to perform data mining and

machine learning analysis (see §3.2.1 to find datasets, at the end of this document for details).

- The group will pitch the project title, identify the datasets, and formulate the project objectives in Week 4 via an assessed short presentation (**Deliverable 1**).
- The project should incorporate both generative and discriminative tasks, showcasing various algorithms discussed in the syllabus that are relevant to the chosen dataset(s).
- The project must include key components such as data preprocessing, model training, performance evaluation, and data visualisation.
- Teams should provide a comprehensive documentation detailing their approach, results, discussions, and analysis.
- (**Campus-dependent**) All documentation, commentary, and code must be hosted on GitHub or Gitlab for Edinburgh (e.g., README/Jupyter Notebook files/Documentation).
- The final three deliverables are i) a 6-page report in pdf format and a fully documented repository of the code accompanying your project; you don't need to submit anything, but you must ensure that all teaching staff have access to it. (**Deliverable 2**); ii) a 15-minute mini-viva consisting of a 5-slide 7-minute presentation followed by a Q&A session conducted by 2 members of staff/lab helpers (**Deliverable 3**), iii) a peer-assessment form using MS Forms (**Deliverable 4 – 0 marks**).
- For deliverables 1-3 the grading is based on a marking rubric (check details on CANVAS).

Group size: 5 people in each group.
Effort: approx. 40+ hours per student
Credit: 40% of overall mark
Coursework timeline: 14th September 2026 (Coursework Released)
Deliverables:

i) **Week 4 – Project Pitch; short presentation to ensure project meets the coursework criteria;**
ii) **Week 11 – 6-page Project Report and Code;**
iii) **Week 12 – Project Presentation; 15-minute mini-viva;**
iv) **Week 12 – Peer Assessment (0 marks); via MS Forms.**

NOTE: Assessment will be continuous throughout course (attendance, discussions, checks).

Deadline Submission (Project Report): 15:30 (local time) on 25th November 2026 (Week 11)

3. Coursework Overview

The group project at hand involves choosing one topic of application and then selecting and analysing 1–3 unique datasets while dividing the project into four distinct phases.

Note that the number of datasets is not prescriptive; just one dataset might be adequate, but in most cases, you will need two (e.g., one tabular and another image dataset). The important thing is to choose datasets that are good candidates to fulfil the coursework requirements, especially regarding modelling: as we will find out throughout the course not all models

perform equally well for all input types present in datasets. The project's overarching goal is to explore, analyse, and apply machine learning techniques to these datasets, contributing valuable insights and solutions to relevant research questions. The project emphasises teamwork, data exploration, and the application of various machine learning methods, including linear classifiers/ regression models, decision trees, and neural networks.

3.1. Topics

Here is a list of quite broad topics; you can use these as a guideline for choosing a domain of application and then proceed to identify suitable dataset(s).

Important Note: Some topics (indicated with an asterisk *) are considered more advanced, hence you should consult with the teaching team first during the labs before committing:

Entertainment

1. Genre prediction of a song based on metadata/visual (album cover)/audio features
2. Popularity prediction of a song based on metadata/visual (album cover)/audio features
3. Genre prediction of a game based on textual description*
4. Movie/Book/Game Review classification*
5. Predict the emotion of a song from audio features and metadata

Natural Sciences

6. Quality prediction of a particular food item or beverage
7. Crop disease prediction from leaf images

Medical

8. Disease diagnosis (e.g., cancer) based on metadata/medical imaging data (e.g., mammograms, radiographs, skin images, retinal images, chest X-rays, MRI scans)

Detection

9. Fake News Detection*, i.e., to detect whether a news article is genuine or misleading
10. Phising email detection*
11. Predict whether a website is fraudulent/scam*
12. Classify images/videos as authentic or manipulated* (DeepFakes)
13. Predict fraudulent insurance claims

3.2. Datasets

Once you have selected a topic, the next step is to select **up to three distinct datasets** (these should be unique to your group project). These datasets will serve as the foundation for the entire project, guiding your project objectives and implementation.

It is imperative you ensure that your dataset(s) are relevant to the topic and can complete the course requirements (see below). You must take into consideration dataset complexities and size when planning your project. We suggest you choose at least one image data set (this will

be applicable for a later part of the course when we talk about CNNs).

Note related to Deliverable 1: Week 4 – Project Pitch. You only need to decide on the datasets to use and formulate the project objectives. You don't need to make any modelling decisions just yet.

3.2.1. Useful links to find datasets

The following list has some pointers to places where you might get some inspiration for data mining challenges together with associated data such as evaluation criteria and comparative performance data

- https://www.kaggle.com - source of lots of different data mining competitions.
- http://www.drivendata.org - source of lots of different data mining competitions with an emphasis on saving the world.
- http://www.kdnuggets.com/competitions/past-competitions.html - list of past data mining competitions; data and evaluation criteria is likely to be available for many of these.
- http://webscope.sandbox.yahoo.com - publicly available research datasets from Yahoo!
- http://www.kdd.org/kdd-cup - KDD Cup is an annual data mining competition run by ACM SIG KDD; datasets, evaluation criteria, and info previous winners are available (note that the most recent competitions are actually hosted on kaggle.com).
- https://huggingface.co/datasets — consolidation of publicly available image, text, and audio datasets
- http://multimediaeval.org/datasets/ - a range of data and evaluation criteria for different types of data mining problems involving multimedia and multi-modal data.

4. Coursework Requirements

Your group project must use your chosen datasets and complete the following requirements (R1–R4). R1 will be assessed both in deliverables D1 and D2-3, whereas the rest in D2 and D3 only (see Section 5 – Deliverables). Note that R2-R4 don't necessarily need to be implemented in this order.

R1. **Project Topic, Direction, and Questions**
In this phase, the group will define a set of clear questions, and objectives based on the selected topic/datasets. Identify the specific problems or hypotheses to be addressed using the datasets (D1-D3).

R2. **Data Analysis and Exploration**
Perform comprehensive data preprocessing and cleaning to ensure data quality. Then apply some exploratory data analysis (EDA) to gain insights into the datasets, including visualisation techniques to present data patterns and trends, and feature selection and evaluation. You are encouraged to also implement a clustering algorithm to show aspects of the data with similar characteristics and evaluating it using the appropriate metrics. Clustering is a great way to interpret and find patterns in the data (D2-D3).

R3. **Baseline Training and Evaluation Experiments**

Apply *at least three (3) machine learning algorithms* to build predictive models, including Decision Trees, Naïve Bayes, Linear Models, Perceptron, k-Nearest Neighbours and any ensemble methods combination. Note your task can be either classification and/or regression, depending on the topic and dataset(s) you choose. Evaluate and compare the models in terms of appropriate metrics for the respective tasks, e.g., accuracy, precision, recall, F-1, error measures (like RMSE) and justify their choice. In the case of Decision Trees discuss their practical applications in the context of the selected datasets/topic (D2-D3).

R4. **Neural Networks**

Implement neural network models, such as Multi-Layer Perceptron (MLPs) and Convolutional Neural Networks (CNN), for tasks such as classification or regression. Train and fine-tune the neural network models. Assess the performance of neural network models and compare them with other techniques. (D2-D3).

Throughout this group coursework, the project team will work collaboratively to navigate through the various requirement phases, from selecting appropriate datasets to applying advanced machine learning techniques. The project's outcomes will contribute to the understanding of the selected datasets, provide solutions to implementation problems, and demonstrate proficiency in data analysis and machine learning. This structured approach will enable you to master essential data science and machine learning skills.

Each group is responsible for project planning, task allocation among team members, and ensuring efficient communication. Regular interactions with instructors and peer code reviews are encouraged to ensure the success of the group projects. Additionally, ethical considerations regarding data usage and privacy must be strictly adhered to when working with real-world datasets.

Finally, it is highly recommended that your dataset(s) satisfies all the above course requirements. If you are unsure whether your choice will satisfy the above requirements, you should ask as soon as possible.

5. Deliverables

There are three deliverables for this coursework:

D1. **Week 4 – Project Pitch**

A 10-minute presentation (we recommend presenting using a short slide-deck, but it's not compulsory). The group will pitch the project title, identify the datasets, formulate the research question and outline any relevant hypotheses. You should also provide a tentative project plan and timeline (deliverables/stages/milestones). You can describe initial thoughts on which approaches they are going to use (or are thinking of taking). There is no requirement to discuss about models yet. This will be arranged via a booked session, where groups will be allocated a time.

D2. **Week 11 – Project Report and Code**

a. A maximum 6-page report in PDF format with the following sections: Introduction,

Related Work, Dataset Description and Analysis, Experimental Setup, Results, Discussion, and Conclusion. Your report should cover all course requirements described in Section 4 – R1-R4, explaining your approach in each part, summarising results in graphs/tables when possible and explaining insightful conclusions drawn. You should also very briefly describe the contribution of each member as an appendix (won't contribute to the total number of pages).

b. A fully documented repository of the code accompanying your project. Your final project should demonstrate a clear understanding of the covered concepts and techniques in data mining and machine learning.

D3. **Week 12 – Project Presentation**
A 15-minute mini-viva comprising a short slide-deck of up to 5 slides (7 minutes of presentation), followed by a Q&A session (8 minutes) conducted by members of staff/lab helpers. This will be your team's chance to present your work and describe approaches you have taken. This will be arranged via a booked session, where groups will be allocated a time. You might be asked to demo your code. **Note all members of the team need to contribute to the presentation and Q&A**.

D4. **Week 12 – Peer Assessment**
This is a MS Form for providing a peer assessment review of your group. This peer review form is done privately and **only** the course lecturers will have access to your answers.

Important notes:

- **Regular attendance in labs is required by all team members** (actively demonstrate and present work during scheduled sessions)
- All group members should contribute equally to the project; it's up to you to decide who gets to do what. You will have the chance to comment on aspects of collaboration and communication, including individual contribution in the Peer Assessment.
- **Any data or resources used for the project must be accessible** (e.g., downloadable/run/test any submission on another machine)
- **Important: Please study the CW Handbook**

If you have any questions or queries about the assessment, please do not hesitate to contact your teaching team.

6. Late Submissions

Late submissions will be subject to the normal penalties as defined in the late coursework policy. The University recognises that, on occasion, students may be unable to submit coursework on the submission date or be unable to present their work on the submission date. In these cases, the University's Submission of Coursework Policy outlines are:

- No individual extensions are permitted under any circumstances.
- Standard 30% deduction from the mark awarded (maximum of five working days).
- In the case where a student submits coursework up to five working days late, and the student has valid mitigating circumstances, the mitigating circumstances policy will apply, and appropriate mitigation will be applied.
- Any coursework submitted after five working days of the set submission date shall be automatically awarded a no grade with no formative feedback provided.

Please contact your Personal Tutor or Counsellor if you are unable to meet the deadlines or need information for Mitigating Circumstances or Temporal Suspensions of Studies.

7. Plagiarism

In any coursework, it is imperative that students uphold the principles of academic integrity and ethical scholarship. Plagiarism, the act of presenting someone else's ideas, work, or words as our own without proper acknowledgment, is strictly prohibited. We are committed to producing original and authentic content, and any external sources, whether they be ideas, data, text, images, or any other material, must be appropriately referenced and acknowledged using the prescribed citation style. This not only ensures the credibility of our work but also demonstrates our respect for the intellectual contributions of others. Together, let us maintain the highest standards of honesty and integrity throughout our collaborative efforts.

https://www.hw.ac.uk/uk/students/studies/examinations/plagiarism.htm

8. Use of Generative AI

On this course you may use GenAI as follows:

- **Framework statement 1: GenAI Language Support**
    - Where English is not your first language you may use GenAI to translate the assessment instructions or other relevant content.
    - You may not use GenAI to translate your assessment answers, you must write them in English as HW requires all work to be completed in English and to be your own work.
    - No GenAI generated content is permitted in your assessment submission; this statement does not include code, see below.
- **Framework statement 4: GenAI-assisted editing and proofreading**
    - You may use GenAI to edit your work e.g. enhancing spelling, grammar, tone, clarity, image / video / audio editing.
- **Framework statement 6: GenAI task completion, human evaluation**

    - You may use GenAI to complete ONLY the task of code generation. You then need to evaluate this output yourself with no GenAI assistance to provide the assessment submission.
- The use of generative AI should be referenced using the appropriate referencing format for the piece of work. If you use it to generate code, you should explicitly say so in comments, *by providing the prompt you used in full*. If you use it to prepare part of your report you should have a clearly outlined disclaimer.
- Any use of artificial intelligence must align with the principles of academic integrity as outlined by the International Centre for Academic Integrity.
- **Note**: Using artificial intelligence to complete a whole assignment for you would not be acting with academic integrity, as you would be presenting work that is not your own. This is plagiarism and would be considered academic misconduct.

9. Resources/Useful Links

The following list has some pointers to places where you might get some inspiration for data mining challenges together with associated data such as evaluation criteria and comparative performance data:

9.1. Getting started with Python

We recommend using Conda or uv to create a virtual environment (with Python >=3.11) for the project.

- Installation: https://docs.conda.io/projects/conda/en/latest/user-guide/install/index.html
- Managing your environment: https://conda.io/projects/conda/en/latest/user-guide/tasks/manage-environments.html#activating-an-environment
- Installation using uv: https://docs.astral.sh/uv/

9.2. Framework recommendations for the course

We recommend the frameworks used in the labs. However, if you want to use others like PyTorch and Hugging Face you should feel free to do so. Remember, documentation is your best friend.

Check your assessment page for further information on Marking

Rubric and submission requirements